

WaNet 论文复现实验报告

论文信息

- 论文题目：WaNet: Imperceptible Warping-based Backdoor Attack
- 作者：Tuan Anh Nguyen, Anh Tuan Tran
- 会议：ICLR 2021
- 论文方向：AI 安全，后门攻击，模型鲁棒性评估
- 官方实现：https://github.com/VinAIResearch/Warping-based_Backdoor_Attack-release
- OpenReview：<https://openreview.net/forum?id=eEn8KTtJOx>

背景：关注什么问题

深度神经网络在图像分类等任务中表现良好，但训练数据和模型训练流程一旦被攻击者污染，模型可能学到隐藏的后门行为。后门攻击的典型目标是：模型在正常输入上保持较高准确率，但在带有特定触发器的输入上输出攻击者指定的标签。

传统后门攻击通常依赖可见的贴片、图案或局部扰动作为触发器。这类触发器虽然容易实现，但也更容易被人工检查、数据增强、异常检测或防御方法发现。WaNet 关注的问题是：能否构造一种视觉上更隐蔽、更难被已有防御发现的后门触发方式。

问题：挑战是什么

WaNet 需要同时满足三个目标：

1. 隐蔽性：触发器不能像固定贴片一样明显，输入图像被攻击后应尽量保持自然。
2. 有效性：后门样本应以较高概率被分类到攻击目标标签，攻击成功率要高。
3. 正常性能：模型在干净测试集上的准确率不能明显下降，否则攻击容易暴露。

复现实验本身也有挑战。该论文的攻击依赖随机形变网格、训练过程中的 poisoned samples 和 noise samples，同时 defense 实验涉及 STRIP、Fine-Pruning、Neural Cleanse 等不同流程。为了避免只复现单一配置，本次实验覆盖了多个数据集和攻击模式。

方法：如何解决挑战

WaNet 的核心思想是使用图像形变作为后门触发器，而不是添加显式贴片。攻击者生成一个平滑的 warping field，对输入图像做轻微空间扭曲，使图像内容整体保持自然，但模型会学习到这种空间形变与目标标签之间的关联。

本次复现使用官方实现，并在本地环境中补充了运行脚本和结果管理逻辑。主要实验设置如下：

- 数据集：MNIST、CIFAR-10、GTSRB

- 攻击模式：all2one、all2all
- 主指标：Clean test accuracy、Attack test accuracy、Noise test accuracy
- Defense：STRIP、Fine-Pruning、Neural Cleanse
- 随机种子：seed 0
- 运行方式：使用 tmux 后台运行，GPU0/GPU1 并行执行任务

核心复现命令如下：

```
cd /home/zilia/project/experiment/wanet
bash scripts/start_weekend_tmux.sh
```

状态检查命令如下：

```
bash scripts/status_weekend.sh
tail -f logs/weekend_latest/summary.tsv
```

实验运行和结果截图

实验使用 tmux 后台运行。最终 CPU、GPU0、GPU1 三个 worker 均正常完成。

下面两张截图来自真实 tmux pane，通过 `tmux capture-pane` 捕获最终运行界面。完整捕获文本保存在 `report_assets/tmux_gpu0_final.txt` 和 `report_assets/tmux_gpu1_final.txt`。

tmux capture-pane: gpu0 final screen

generated from real captured text; full text: report_assets/tmux_gpu0_final.txt

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Updated !!!
  Result: Accuracy: 99.100 | Cross Entropy Loss: 0.024773 | Reg Loss: 34.305248
| Reg best: 34.305248
10 labels found
... clipped in PNG, full text saved in report_assets/tmux_gpu0_final.txt ...
```

tmux capture-pane: gpu1 final screen

generated from real captured text; full text: report_assets/tmux_gpu1_final.txt

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Updated !!!
  Result: Accuracy: 98.710 | Cross Entropy Loss: 0.036913 | Reg Loss: 41.006596
| Reg best: 41.006596
10 labels found
... clipped in PNG, full text saved in report_assets/tmux_gpu1_final.txt ...
```

下面截图来自真实运行日志 `logs/weekend_20260613_194939/summary.tsv` 的尾部，记录了训练、评估和防御任务完成时间。

file: logs/weekend_20260613_194939/summary.tsv (tail)

generated from real captured text; full text: report_assets/summary_tail.txt

```
2026-06-13 22:32:33[gpu1]train_mnist_all2all[start_attempt_1]
2026-06-14 00:55:58[gpu0]train_cifar10_all2one[done]
2026-06-14 00:55:58[gpu0]eval_cifar10_all2one[start_attempt_1]
2026-06-14 00:56:05[gpu0]eval_cifar10_all2one[done]
2026-06-14 00:56:05[gpu0]train_cifar10_all2all[start_attempt_1]
2026-06-14 01:32:17[gpu1]train_mnist_all2all[done]
2026-06-14 01:32:17[gpu1]eval_mnist_all2all[start_attempt_1]
2026-06-14 01:32:22[gpu1]eval_mnist_all2all[done]
2026-06-14 01:32:22[gpu1]strip_mnist_all2all[start_attempt_1]
2026-06-14 01:32:55[gpu1]strip_mnist_all2all[done]
2026-06-14 01:32:55[gpu1]fine_pruning_mnist_all2all[start_attempt_1]
2026-06-14 01:34:04[gpu1]fine_pruning_mnist_all2all[done]
2026-06-14 01:34:05[gpu1]strip_cifar10_all2one[start_attempt_1]
2026-06-14 01:35:14[gpu1]strip_cifar10_all2one[done]
2026-06-14 06:03:58[gpu0]train_cifar10_all2all[done]
2026-06-14 06:03:58[gpu0]eval_cifar10_all2all[start_attempt_1]
2026-06-14 06:04:05[gpu0]eval_cifar10_all2all[done]
2026-06-14 06:04:05[gpu0]train_gtsrb_all2one[start_attempt_1]
2026-06-14 06:04:17[gpu1]strip_cifar10_all2all[start_attempt_1]
2026-06-14 06:05:23[gpu1]strip_cifar10_all2all[done]
2026-06-14 06:05:23[gpu1]fine_pruning_cifar10_all2one[start_attempt_1]
2026-06-14 06:18:10[gpu1]fine_pruning_cifar10_all2one[done]
2026-06-14 06:18:10[gpu1]fine_pruning_cifar10_all2all[start_attempt_1]
2026-06-14 06:31:01[gpu1]fine_pruning_cifar10_all2all[done]
2026-06-14 10:15:25[gpu0]train_gtsrb_all2one[done]
2026-06-14 10:15:25[gpu0]eval_gtsrb_all2one[start_attempt_1]
2026-06-14 10:15:32[gpu0]eval_gtsrb_all2one[done]
2026-06-14 10:15:32[gpu0]train_gtsrb_all2all[start_attempt_1]
2026-06-14 10:16:03[gpu1]strip_gtsrb_all2one[start_attempt_1]
2026-06-14 10:17:04[gpu1]strip_gtsrb_all2one[done]
2026-06-14 14:26:33[gpu0]train_gtsrb_all2all[done]
2026-06-14 14:26:33[gpu0]eval_gtsrb_all2all[start_attempt_1]
2026-06-14 14:26:40[gpu0]eval_gtsrb_all2all[done]
2026-06-14 14:26:40[gpu0]fine_pruning_gtsrb_all2one[start_attempt_1]
2026-06-14 14:27:07[gpu1]strip_gtsrb_all2all[start_attempt_1]
2026-06-14 14:28:08[gpu1]strip_gtsrb_all2all[done]
2026-06-14 14:28:08[gpu1]fine_pruning_gtsrb_all2one[skipped_exists]
2026-06-14 14:28:08[gpu1]fine_pruning_gtsrb_all2all[start_attempt_1]
2026-06-14 14:42:33[gpu0]fine_pruning_gtsrb_all2one[done]
2026-06-14 14:42:33[gpu0]fine_pruning_gtsrb_all2all[skipped_exists]
2026-06-14 14:42:33[gpu0]neural_cleanse_mnist_all2one[start_attempt_1]
2026-06-14 14:43:29[gpu1]fine_pruning_gtsrb_all2all[done]
2026-06-14 14:43:29[gpu1]neural_cleanse_cifar10_all2one[start_attempt_1]
2026-06-14 15:29:03[gpu0]neural_cleanse_mnist_all2one[done]
2026-06-14 16:10:37[gpu1]neural_cleanse_cifar10_all2one[done]
```

主实验结果截图来自 6 个真实 eval_results.json 文件，路径为 checkpoints/<dataset>/<attack_mode>/eval_results.json 。

files: checkpoints/*/eval_results.json

generated from real captured text; full text: report_assets/eval_json_files.txt

```
===== checkpoints/mnist/all2one/eval_results.json =====
{
  "clean_acc": 99.5,
  "bd_acc": 99.97999572753906,
  "cross_acc": 99.15999603271484
}
===== checkpoints/mnist/all2all/eval_results.json =====
{
  "clean_acc": 99.44999694824219,
  "bd_acc": 99.15999603271484,
  "cross_acc": 99.18000030517578
}
===== checkpoints/cifar10/all2one/eval_results.json =====
{
  "clean_acc": 94.5199966430664,
  "bd_acc": 99.3699951171875,
  "cross_acc": 92.5199966430664
}
===== checkpoints/cifar10/all2all/eval_results.json =====
{
  "clean_acc": 94.36000061035156,
  "bd_acc": 93.5199966430664,
  "cross_acc": 92.64999389648438
}
===== checkpoints/gtsrb/all2one/eval_results.json =====
{
  "clean_acc": 99.34283447265625,
  "bd_acc": 99.74663543701172,
  "cross_acc": 99.02613067626953
}
===== checkpoints/gtsrb/all2all/eval_results.json =====
{
  "clean_acc": 99.5882797241211,
  "bd_acc": 99.22406768798828,
  "cross_acc": 99.14488983154297
}
```

STRIP 防御结果截图来自 defenses/STRIP/results/ 下的真实输出文件。由于原文件每行包含 1000 个 entropy 数值，截图中展示每个文件的行数、样本数以及前缀内容。

files: defenses/STRIP/results/*/*_output.txt

generated from real captured text; full text: report_assets/strip_file_snippets.txt

===== defenses/STRIP/results/mnist/all2one/mnist_all2one_output.txt =====

line count: 1

word count line1/line2: 1000 1000

line1 prefix: 1.7567672729492188 1.5531467199325562 1.9485747814178467 1.9212007522583008 1.4663206338882446 1.973294734954533214569092 1.6552467346191406 1.5374606847763062 1.49244141 ...
line2 prefix: 1.968085527420044 1.7228606939315796 2.077157497406006 1.8519318103790283 1.5865837335586548 2.03639841079718932762146 1.901411771774292 1.5867770910263062 1.627685904502 ...

===== defenses/STRIP/results/mnist/all2all/mnist_all2all_output.txt =====

line count: 1

word count line1/line2: 1000 1000

line1 prefix: 2.2726223468780518 2.862697124481201 2.0969765186309814 2.533834457397461 2.1422646045684814 2.250901460647542055511475 2.2207131385803223 2.3161911964416504 1.8391109704 ...
line2 prefix: 2.0066680908203125 2.3326940536499023 2.1306302547454834 2.4403235912323 1.6986764669418335 2.1632583141326913139343262 1.9642959833145142 1.6844874620437622 1.8115283250 ...

===== defenses/STRIP/results/cifar10/all2one/cifar10_all2one_output.txt =====

line count: 1

word count line1/line2: 1000 1000

line1 prefix: 2.816593885421753 2.5110785961151123 2.6743602752685547 2.7138330936431885 2.6615381240844727 2.6385114192965799083709717 2.6127681732177734 2.501107931137085 2.477800607 ...
line2 prefix: 2.627016067504883 2.3843958377838135 2.549948215484619 2.5964086055755615 2.554051399230957 2.511522293090823303833008 2.656672716140747 2.585564613342285 2.3773021697998 ...

===== defenses/STRIP/results/cifar10/all2all/cifar10_all2all_output.txt =====

line count: 1

word count line1/line2: 1000 1000

line1 prefix: 2.8154778480529785 2.316180944442749 2.628955125808716 2.5155272483825684 2.7716381549835205 2.700856924057085816955566 2.7960779666900635 2.565347909927368 2.70502662658 ...
line2 prefix: 2.6504952907562256 2.2375988960266113 2.535649299621582 2.5196404457092285 2.59989070892334 2.559971332550044130096436 2.72190523147583 2.466970920562744 2.52112507820129 ...

===== defenses/STRIP/results/gtsrb/all2one/gtsrb_all2one_output.txt =====

line count: 1

word count line1/line2: 1000 1000

line1 prefix: 13.572887420654297 11.063027381896973 12.378273010253906 13.2748441696167 12.244919776916504 13.124176979064930274963379 11.406442642211914 13.276684761047363 11.51723384 ...
line2 prefix: 10.734365463256836 10.786886215209961 11.003015518188477 9.810818672180176 11.156198501586914 11.34094619750548240661621 11.962477684020996 10.638051986694336 11.53976345 ...

===== defenses/STRIP/results/gtsrb/all2all/gtsrb_all2all_output.txt =====

line count: 1

word count line1/line2: 1000 1000

line1 prefix: 12.359081268310547 13.191291809082031 14.020638465881348 12.878362655639648 13.061850547790527 13.911345481802069091796875 13.337421417236328 12.489662170410156 13.261396 ...
line2 prefix: 11.967864036560059 13.022177696228027 12.8540678024292 12.405078887939453 12.58474349975586 13.181955337524443043518066 13.390399932861328 11.510417938232422 12.973903656 ...

Fine-Pruning 防御结果截图来自 defenses/fine_pruning/results/ 下的真实结果文件，展示每个文件的开头和结尾部分。

files: defenses/fine_pruning/results//*_results.txt**

generated from real captured text; full text: report_assets/fine_pruning_file_snippets.txt

===== defenses/fine_pruning/results/mnist/all2one/mnist_all2one_results.txt =====

--- head ---

0 99.5000 99.9800
1 99.4800 99.9600
2 99.3700 99.9900
3 99.2800 99.9900
4 99.3300 99.9800

--- tail ---

59 23.6500 99.9200
60 35.4200 92.0600
61 33.6900 79.8400
62 19.7100 91.4900
63 14.3800 90.3500

===== defenses/fine_pruning/results/mnist/all2all/mnist_all2all_results.txt =====

--- head ---

0 99.4500 99.1600
1 99.3700 99.1400
2 99.4600 99.0500
3 99.4200 98.8700
4 99.3500 98.8600

--- tail ---

59 52.1800 61.5400
60 54.7200 51.9400
61 43.8600 41.9400
62 32.9200 22.4900
63 23.5000 16.1700

===== defenses/fine_pruning/results/cifar10/all2one/cifar10_all2one_results.txt =====

--- head ---

0 94.5200 99.3700
1 94.5300 99.3200
2 94.5100 99.2900
3 94.4600 99.2700
4 94.4600 99.2700

--- tail ---

507 42.0500 99.9000
508 30.6700 99.9000
509 29.8200 99.8400
510 21.2100 99.6900
511 16.6100 89.7000

===== defenses/fine_pruning/results/cifar10/all2all/cifar10_all2all_results.txt =====

--- head ---

0 94.3600 93.5200
1 94.3900 93.5200
2 94.3600 93.5200
3 94.3600 93.5300
4 94.3500 93.5400

--- tail ---

507 47.1700 53.0600
508 51.3400 49.9500
509 34.7700 32.2000
510 27.8700 25.7400
511 19.0900 19.2600

===== defenses/fine_pruning/results/gtsrb/all2one/gtsrb_all2one_results.txt =====

--- head ---

0 99.3428 99.7466
1 99.3508 99.7466
2 99.3508 99.7387
3 99.3508 99.7387
4 99.3349 99.7387

--- tail ---

507 34.0301 99.4933
508 27.1338 99.6279
509 16.5321 99.7150
510 7.7831 99.6991
511 7.2605 99.9842

```
===== defenses/fine_pruning/results/gtsrb/all2all/gtsrb_all2all_results.txt =====
--- head ---
0 99.5883 99.2241
1 99.5883 99.2082
2 99.5883 99.2082
3 99.5883 99.2161
4 99.5883 99.2003
--- tail ---
507 43.7055 34.8456
508 28.5273 22.0190
509 26.9596 21.5835
510 22.5178 11.1481
511 8.8044 6.6033
```

Neural Cleanse 结果截图来自 defenses/neural_cleanse/results/ 下的真实输出文件。本次只运行了 MNIST all2one 和 CIFAR-10 all2one 两组 Neural Cleanse。

```
files: defenses/neural_cleanse/results/*/**_output.txt
generated from real captured text; full text: report_assets/neural_cleanse_file_tail.txt

===== defenses/neural_cleanse/results/mnist/all2one/all2one_mnist_output.txt =====
34.333412, 3.1642964, 3.2304192
39.46388, 68.30318, 24.111408, 30.75564, 35.440678, 33.991707, 35.16794, 40.258858, 34.333412, 32.199123
33.992214, 3.4686394, 2.9542453
36.872173, 72.76343, 23.745003, 30.558146, 36.10801, 32.83867, 33.992214, 40.778675, 35.211708, 31.652649
34.30166, 4.8116884, 2.115634
38.556946, 65.896614, 24.121887, 30.580877, 35.80166, 31.05622, 34.30166, 41.641495, 34.600018, 32.28357
32.88228, 3.6548438, 2.3733764
38.16626, 66.46265, 24.207958, 30.41712, 35.452393, 31.521759, 34.388206, 40.76574, 32.88228, 31.962334

===== defenses/neural_cleanse/results/cifar10/all2one/all2one_cifar10_output.txt =====
38.6582, 7.238155, 1.1904981
30.04119, 32.821228, 54.798416, 40.671875, 38.12772, 33.77613, 38.6582, 50.93956, 45.842484, 39.20143
37.27381, 4.4332194, 1.6667982
29.88453, 34.469894, 57.732647, 36.497322, 37.27381, 40.263977, 34.112007, 51.99245, 46.48501, 39.0316
39.736347, 3.4524302, 2.6571789
30.562622, 34.055603, 58.99507, 40.93976, 37.407715, 41.74185, 39.736347, 52.303123, 44.989426, 37.566925
39.131172, 4.8716526, 1.0729609
42.399853, 34.973152, 58.93653, 39.13636, 37.68061, 35.845287, 33.90408, 49.675232, 45.819866, 39.131172
```

结果：效果如何

主实验结果

all2one 参考值来自官方 README 的 Results 表；all2all 参考值使用论文/复现整理中的对应参考值。官方 README 说明 Noise test 可能由于随机噪声生成而存在波动。

数据集	攻击模式	复现 Clean	复现 Attack	复现 Noise	参考 Clean	参考 Attack	参考 Noise	差异 Clean	差异 Attack	差异 Noise
MNIST	all2one	99.50	99.98	99.16	99.52	99.86	98.20	-0.02	+0.12	+0.96
CIFAR-10	all2one	94.52	99.37	92.52	94.15	99.55	93.55	+0.37	-0.18	-1.03
GTSRB	all2one	99.34	99.75	99.03	98.87	99.33	98.01	+0.47	+0.42	+1.02

数据集	攻击模式	复现 Clean	复现 Attack	复现 Noise	参考 Clean	参考 Attack	参考 Noise	差异 Clean	差异 Attack	差异 Noise
MNIST	all2all	99.45	99.16	99.18	99.44	95.90	94.34	+0.01	+3.26	+4.84
CIFAR-10	all2all	94.36	93.52	92.65	94.43	93.36	91.47	-0.07	+0.16	+1.18
GTSRB	all2all	99.59	99.22	99.14	99.39	98.31	98.96	+0.20	+0.91	+0.18

总体来看，6 个主设置均成功复现出高 clean accuracy 和高 attack accuracy。all2one 的三组结果与官方表基本一致；CIFAR-10 all2one 的 Noise accuracy 比参考值低约 1.03 个百分点，但仍在可接受波动范围内。MNIST all2all 的 Attack 和 Noise 指标高于参考值，说明本次训练得到的后门更强，但 clean accuracy 与参考值保持一致。

STRIP 结果

STRIP 在本次复现中的检测阈值为 0.2。6 个设置的 trojan entropy 最小值均显著高于 0.2，因此 STRIP 不会将这些样本稳定判定为后门样本。这与论文中 WaNet 能绕过 STRIP 的结论一致。

数据集	攻击模式	Trojan entropy 最小值	Trojan entropy 均值	Benign entropy 均值
MNIST	all2one	1.177	1.703	1.772
MNIST	all2all	1.734	2.295	1.916
CIFAR-10	all2one	2.216	2.684	2.637
CIFAR-10	all2all	2.019	2.761	2.700
GTSRB	all2one	9.270	11.999	10.588
GTSRB	all2all	10.189	12.804	12.565

Fine-Pruning 结果

Fine-Pruning 的趋势也与论文结论一致：当 clean accuracy 仍然保持在 90% 以上时，attack accuracy 通常仍然较高。这说明 WaNet 后门并不容易通过简单剪枝完全消除。

数据集	攻击模式	原始 Clean / Attack	Clean 仍大于 90% 时的最后结果
MNIST	all2one	99.50 / 99.98	90.69 / 99.91
MNIST	all2all	99.45 / 99.16	90.22 / 95.97
CIFAR-10	all2one	94.52 / 99.37	90.02 / 99.91
CIFAR-10	all2all	94.36 / 93.52	90.36 / 86.74
GTSRB	all2one	99.34 / 99.75	90.27 / 99.76

数据集	攻击模式	原始 Clean / Attack	Clean 仍大于 90% 时的最后结果
GTSRB	all2all	99.59 / 99.22	90.35 / 70.38

Neural Cleanse 结果

Neural Cleanse 仅复现了 MNIST all2one 和 CIFAR-10 all2one。官方实现说明 Neural Cleanse 本身不稳定，anomaly index 可能随运行变化。本次复现中 anomaly index 没有形成持续稳定的强检测信号，整体符合 WaNet 难以被 Neural Cleanse 稳定检出的趋势。

复现范围说明

本次已完成：

- MNIST、CIFAR-10、GTSRB 三个数据集的 all2one 和 all2all 主结果
- STRIP：3 个数据集 × 2 个攻击模式
- Fine-Pruning：3 个数据集 × 2 个攻击模式
- Neural Cleanse：MNIST all2one、CIFAR-10 all2one

本次未覆盖：

- CelebA 实验
- 多随机种子平均和方差
- 论文中的所有消融实验
- 所有可视化与感知性分析
- Neural Cleanse 在所有数据集和攻击模式上的完整组合

结论

本次复现实验成功复现了 WaNet 的主要攻击效果。模型在干净样本上保持较高准确率，同时在后门样本上具有较高攻击成功率，说明基于图像空间形变的后门触发器能够兼顾隐蔽性和攻击有效性。防御实验也显示，STRIP 和 Fine-Pruning 难以稳定消除或检测 WaNet 后门，Neural Cleanse 的检测结果也不稳定。这与论文对 WaNet 隐蔽性和防御绕过能力的分析基本一致。

成员贡献

- 周争妍：完成论文选择与资料整理；配置本地 Conda/GPU 实验环境；修复官方代码在当前环境中的兼容性和结果污染问题；编写 tmux 后台运行脚本；完成 MNIST、CIFAR-10、GTSRB 上 all2one/all2all 主实验；完成 STRIP、Fine-Pruning、Neural Cleanse 防御实验；整理实验结果、截图和报告。
- 成员 2：